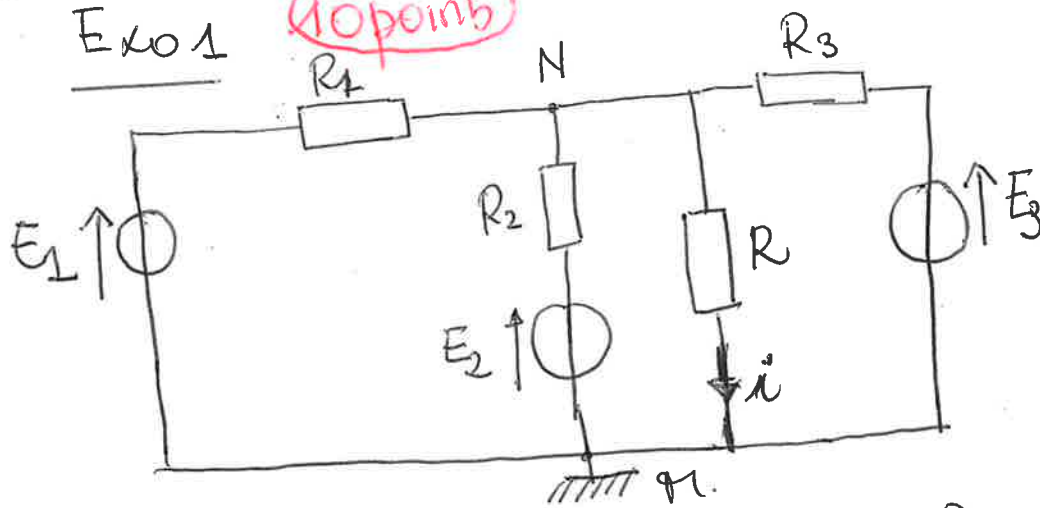


Bonème Examen Electrocinétique
L1 TECHNO 2024/2025
Session 1

1/5

Exo 1

10 points



Determinons i qui circule dans R par :

1) Théorème de Millmann

$$V_N = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{0}{R} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R} + \frac{1}{R_3}} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R} + \frac{1}{R_3}} \quad (2)$$

$V_N = 4,6V$

or $i = \frac{V_N - V_{\pi}}{R} = \frac{V_N}{R}$

AN: $i = \frac{4,6}{2} = 2,3A$

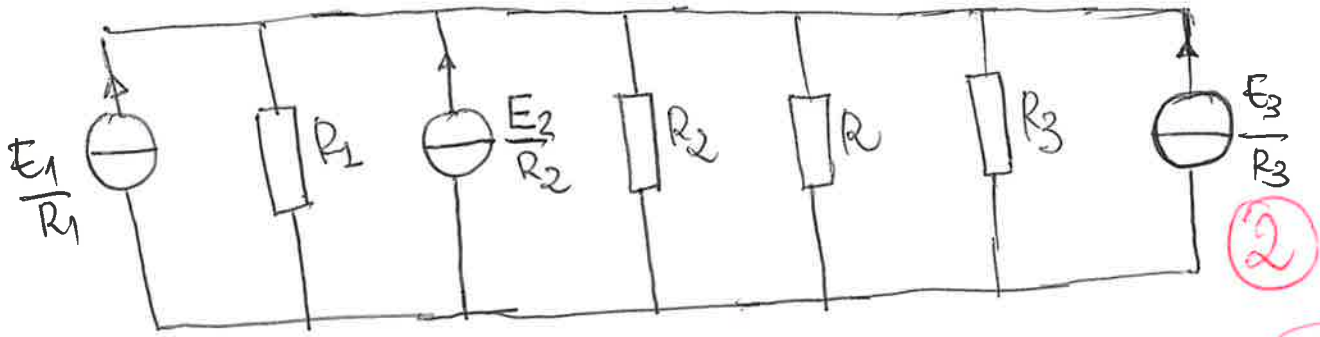
$i = 2,3A$

1

2) Méthode des équivalences des générateurs Thévenin $\leftrightarrow$ Norton

Transformation des générateurs Thévenin → Norton

(2/5)

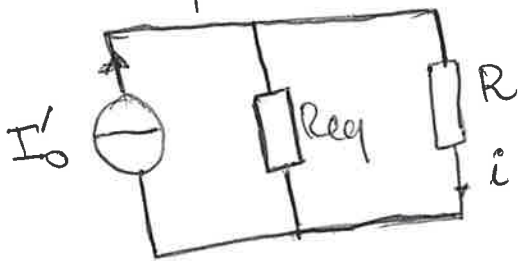


$$I_0 = \frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}$$

AN: $I_0' = 5,75 \text{ A}$

$$R_{eq} = R_1 \parallel R_2 \parallel R_3$$

AN: $R_{eq} = \frac{2}{3} R = \frac{4}{3} \Omega \approx 1,33 \Omega$



Par le PDC:

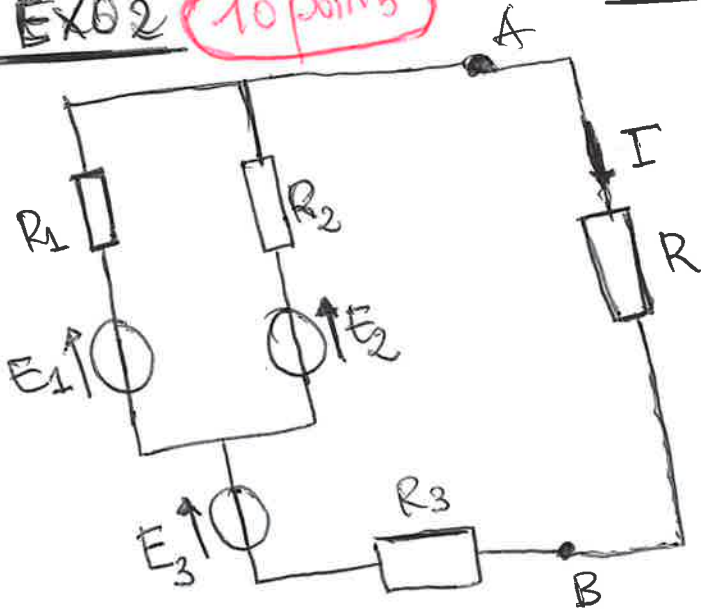
$$i = I_0' \frac{R_{eq}}{R + R_{eq}}$$

AN: $i = 2,3 \text{ A}$

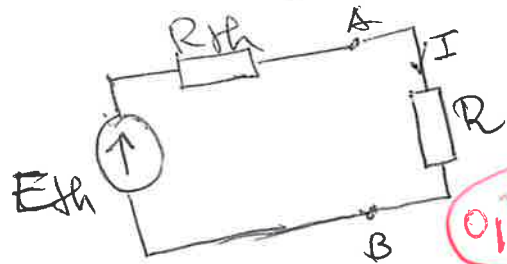
EX02

10 points

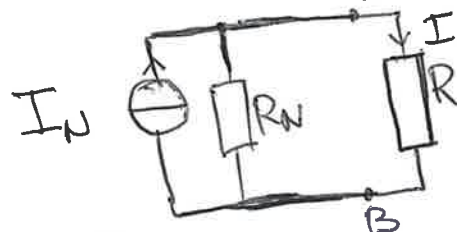
EX02



Schema équivalent de Thévenin

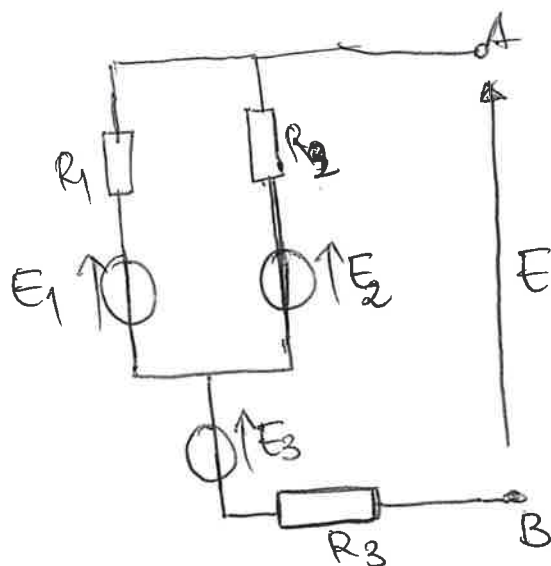


Schema équivalent de Norton

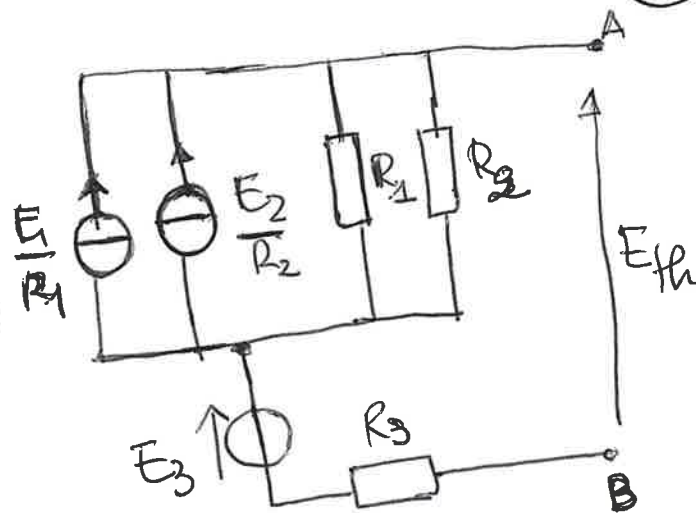


Calcul de E_{th} .

3/5



0,5



Le courant électromoteur équivalent I_0

$$I_0 = \frac{E_1}{R_1} + \frac{E_2}{R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

AN: $I_0 = 2A$

AN: $R_{eq} = 1\Omega$

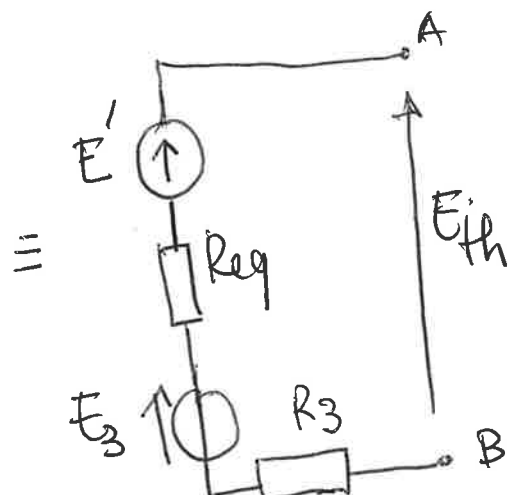
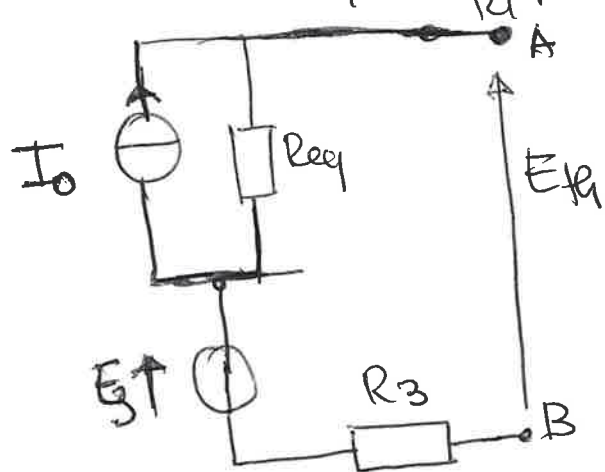


Fig 1

$$E' = R_{eq} I_0 = \frac{R_2 E_1 + R_1 E_2}{R_1 + R_2}$$

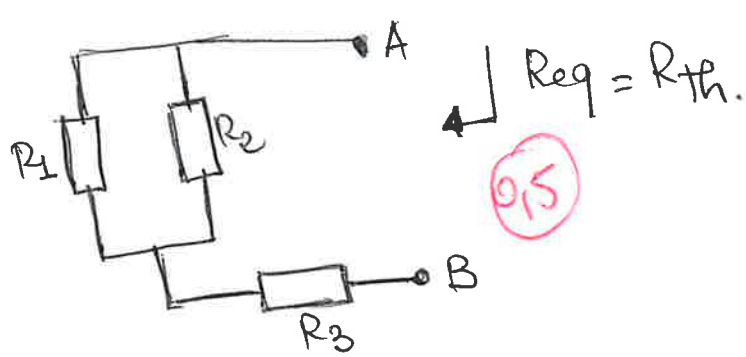
AN: $E' = 2V$

$$E_{th} = E' + E_3$$

$\Rightarrow$ AN: $E_{th} = 4V$

1

+ 2 pts pour la démarche

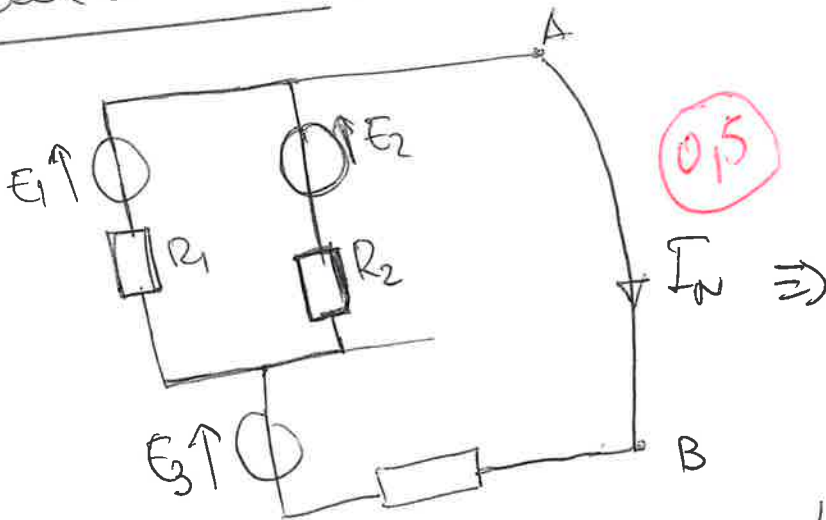
R_{TH}

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

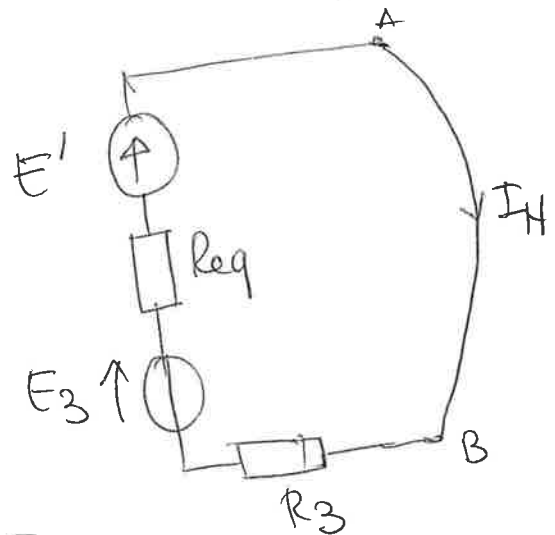
$$R_{TH} = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 + R_2}$$

AN: $R_{TH} = 3 \Omega$

1

calcul de I_N

A partir Fig. ①
de E_{TH} on
obtient

Par Poullet:

$$I_N = \frac{E' + E_3}{R_{eq} + R_3}$$

AN:

$I_N = \frac{4}{3} A \approx 1,33 A$

Calcul de R_N

$R_N = R_{TH} = 3 \Omega$

0,5

2) d'intensité du courant I traversant la résistance R . (5/5)

Méthode de Thevenin

$$I = \frac{E_{th}}{R_{th} + R}$$

so:

$$I = 0,5 \text{ A}$$

0,5

Méthode de Norton

$$I = I_N \cdot \frac{R_N}{R_N + R}$$

AN:

$$I = 0,5 \text{ A}$$

3) Puissance délivrée par le réseau d'alimentation

$$P_f = E_{th} I - R_{th} I^2$$

$$P_f = 1,25 \text{ W}$$

ou

$$P_{dissipée} = R I^2$$

$$P_{diss} = 1,25 \text{ W}$$